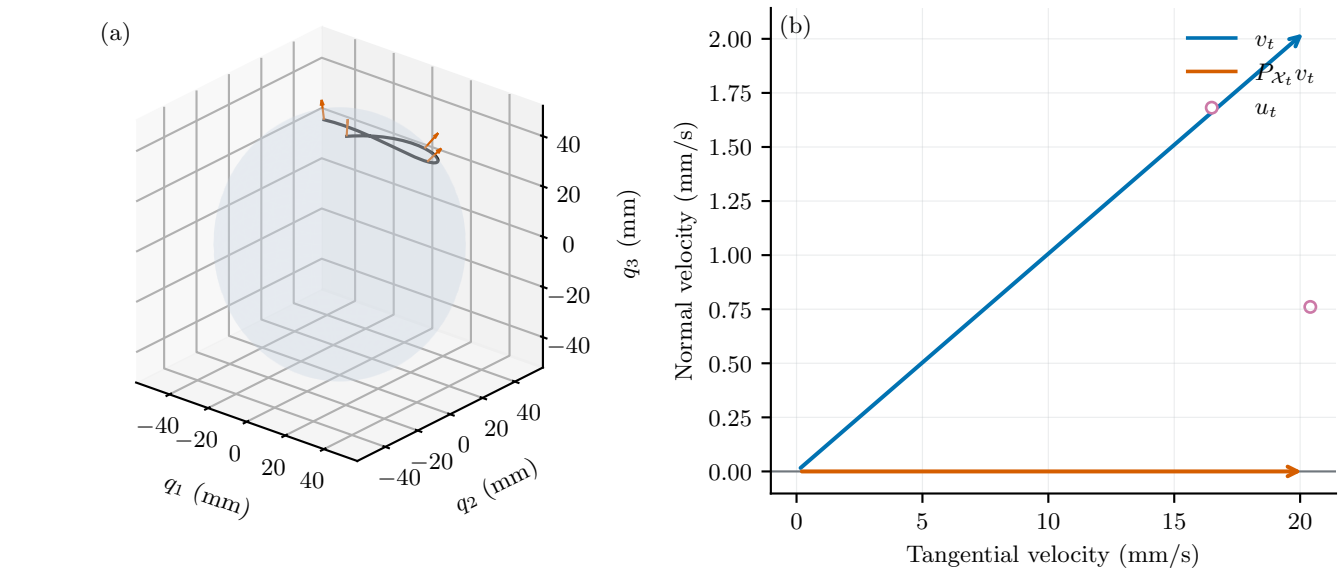

(a)
40
20
0
−20
−40
q_3 (mm)
−40
−20
0
20
40
q_1 (mm)
−40
−20
0
20
40
q_2 (mm)
(b)
v_t
$P_{\mathcal{X}_t} v_t$
u_t
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Normal velocity (mm/s)
0
5
10
15
20
Tangential velocity (mm/s)